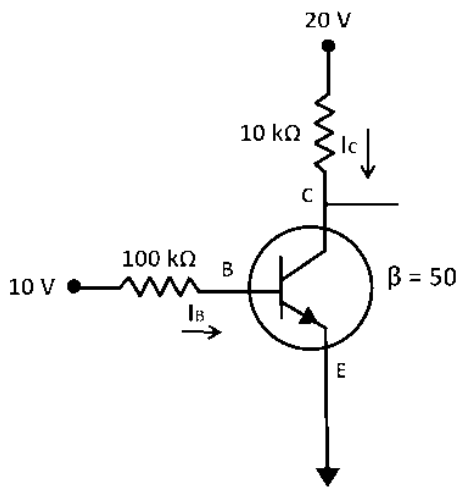


BJT Basics

Review Exercise 1



If the given BJT is in forward active mode-

- Find I_B .
- Find I_C .
- Find V_C .
- What verifies that the BJT is in forward active mode?

Ans: a) 0.093 mA
b) 4.65 mA
c) -26.5 V
d) $V_E > V_C$ or $V_{CE} > 0$

Review Practice Problem 1

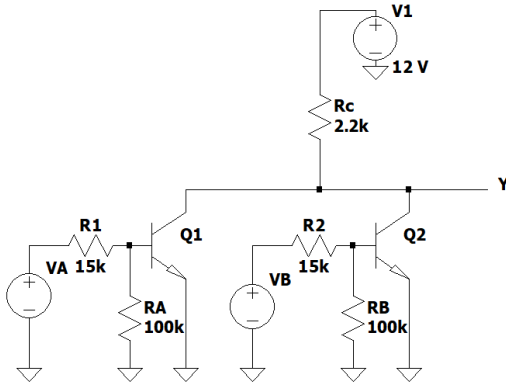
If the BJT in Review Exercise 1 is in saturation mode-

- Find I_B .
- Find I_C .
- Find I_E .

Ans: a) 0.092 mA
b) 1.98 mA
c) 2.072 mA

Basic Operation

Exercise 1



For the RTL NOR gate-

- Determine the output voltage V_Y for all logic cases & make a table. Verify your operating mode assumptions.
- What is the high & low threshold for output voltage?
- If R_B is doubled, find the new base current of Q_2 . Will it still satisfy saturation mode conditions for Q_2 for the cases (1,1) & (0,1)?

[Assume 0.2 V as low input voltage, $\beta_F = 30$]

Ans: a) 12 V, 0.2 V, 0.2 V, 0.2 V
b) 12 V, 0.2 V
c) 0.742 mA, Yes

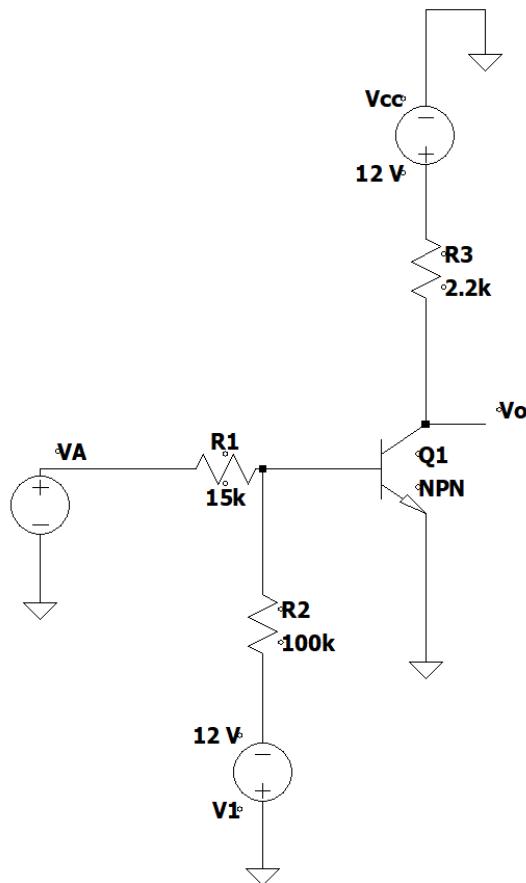
Practice Problem 1:

For the RTL NOR gate in **Exercise 1**, using the value of R_B in (d), find the base voltage of Q_2 and verify whether it meets the condition for cutoff for (0,0) & (1,0).

Ans: $V_B = 0.18 \text{ V} > 0.5 \text{ V}$, satisfies

Power Dissipation

Exercise 2



For the RTL inverter, find the maximum & average power dissipation.

[High input = 12 V, Low input = 0.2 V]

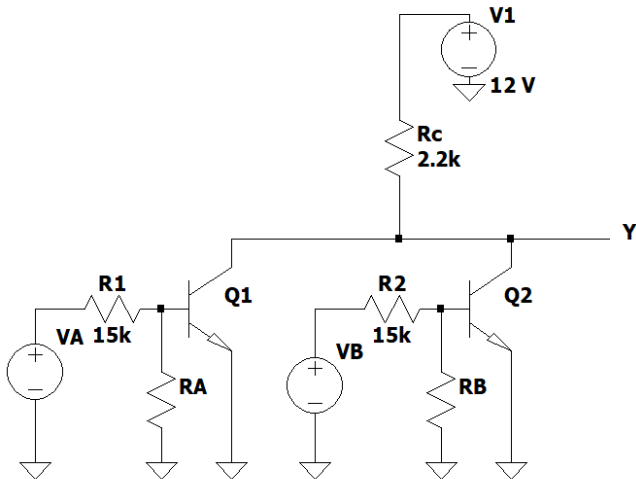
Ans: 82.7782 mW, 42.0362 mW

Practice Problem 2:

For the RTL inverter in **Exercise 2**, find the ratio of maximum & minimum power dissipation, if $V_1(-12\text{ V})$ was missing.

Ans: 212494.7:1

Exercise 3



For the RTL NOR gate shown-

- a) If, $R_A = R_B$, and the power dissipation for logic case (0,0) is $0.25 \mu W$, find the value of the resistances.

$[V_{BE}(\text{saturation}) =$

$0.8 V, V_{CE}(\text{saturation}) = 0.2 V]$

Assume a low input voltage of $0.2 V$

Practice Problem 3:

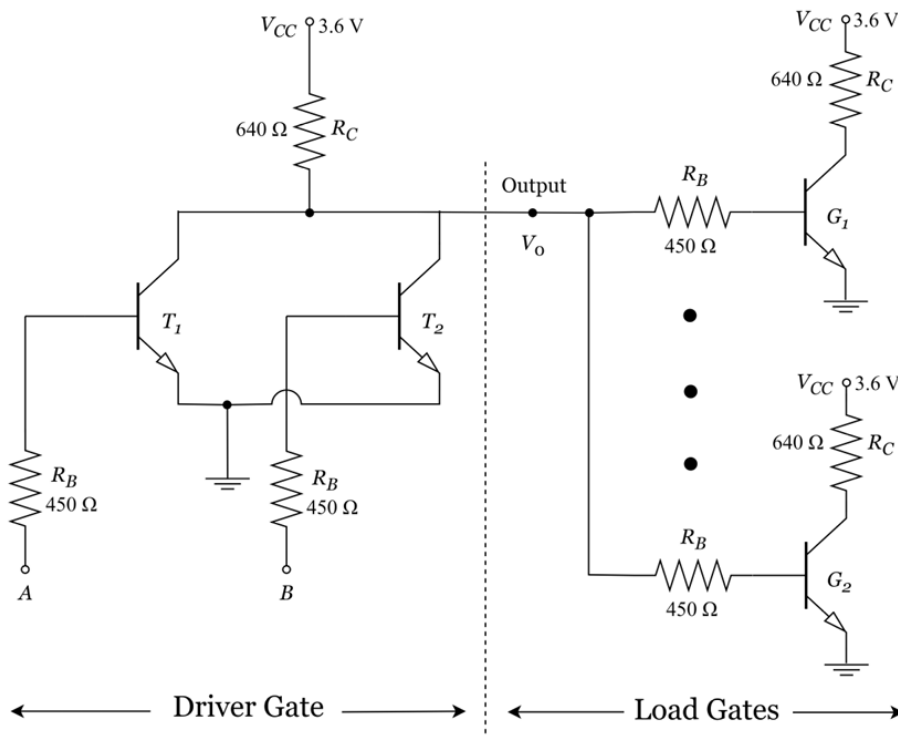
Ans: 145 k Ω

In the circuit of Exercise 3, find a new value of R_C required to double the maximum power dissipation in the circuit.

Ans: 0.966 k Ω

Noise Margin

Exercise 4



For the RTL NOR driver & RTL NOT loads, find the noise margin.

Assume

$$V_{OH} = 3.5 \text{ V}$$

$$V_{IL} = 0.2 \text{ V}$$

$$\beta_F = 30$$

Ans: 0.3 V

Practice Problem 4:

In the circuit of **Exercise 4**, what value of R_B will make $N_H = N_L$?

Ans: 13.55 kΩ

Fanout

Exercise 5

For the RTL NOR driver in **Exercise 4** if $V_{OH} = 1.3 \text{ V}$ -

- Find maximum fanout.
- Find the value of V_O if fanout (N) = 5, and both inputs are low.
- Find $(\beta_F)_{min}$ (for the loads) the power dissipation of the loads only for conditions in (b).
- Find the power dissipation in the driver only when both inputs are high.
- If N=1, what logic function does the driver-load combination implement?

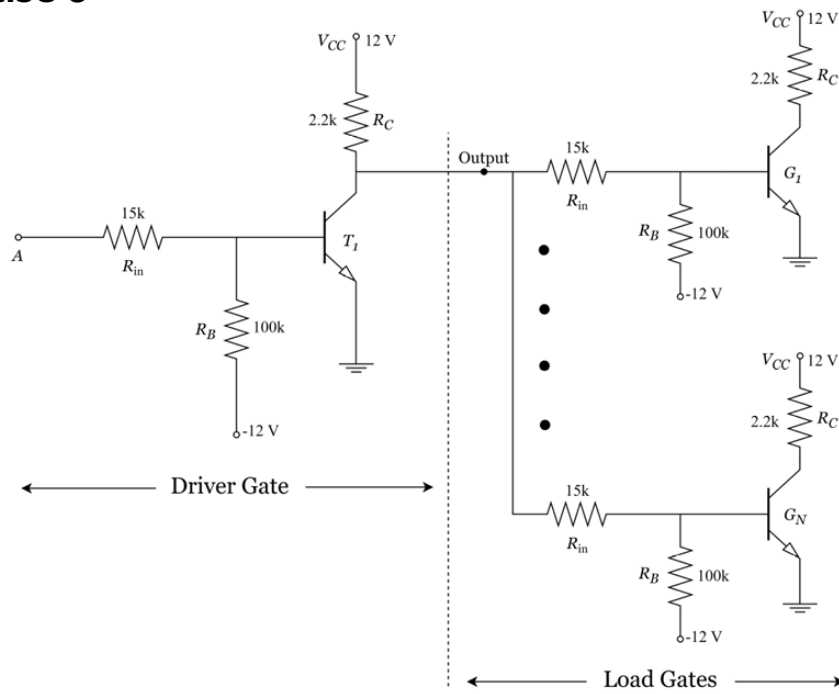
Practice Problem 5:

What value of V_{OH} would reduce the maximum fanout of the RTL NOR gate to 2 in Exercise 5?

$\text{Ans: } 2.18 \text{ V}$

Mixed

Exercise 6



For the given RTL inverters,

$$V_{OH} = 10 \text{ V}, V_{OL} = 0.2 \text{ V}$$

$$\beta_F = 30$$

- Find maximum fanout.
- Find V_o for $N=2$ loads, and the input of the driver is low.
- If V_{in} is high, find the power dissipation of the driver circuit.
- If V_{in} is low and fanout=2, find the power dissipation of the total circuit (driver + loads).
- Find the noise margin.

$e) 0.375 \text{ V}$

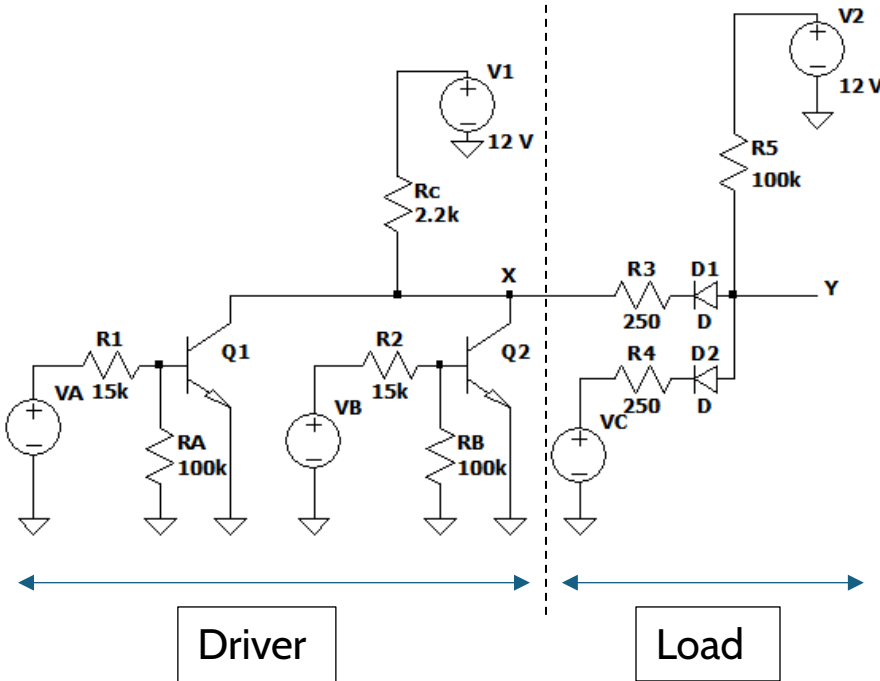
$d) 146.95 \text{ mW}$

$c) 74.86 \text{ mW}$

$b) 9.46 \text{ V}$

$\text{Ans: } a) 1$

Exercise 7



- For the circuit given-
- What logic function does it implement?
 - Find maximum fanout.
 - Find the max power dissipation of the circuit.

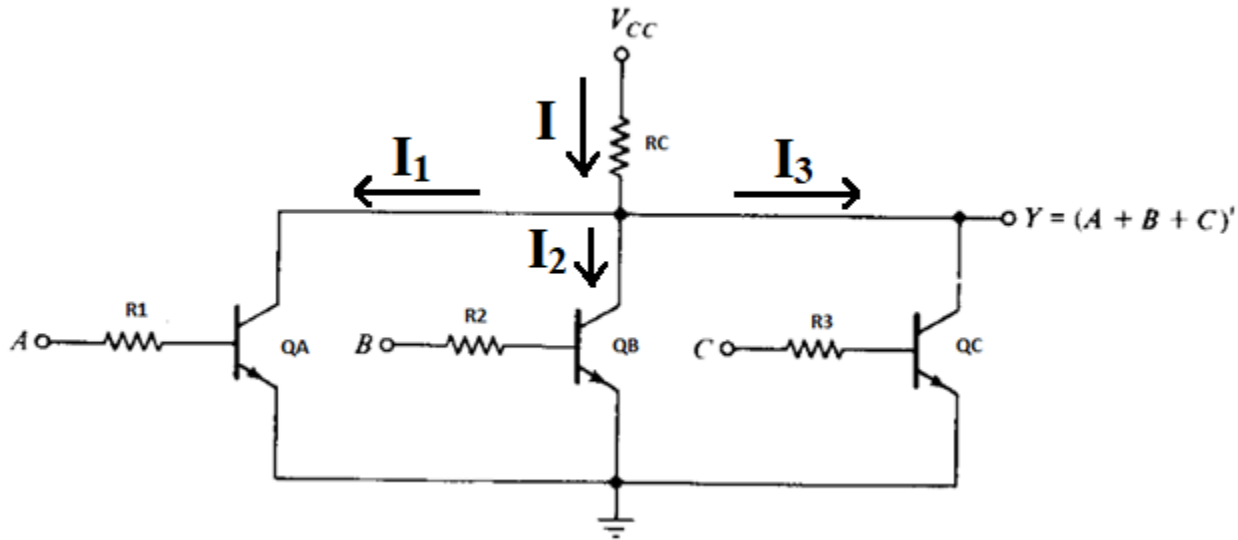
Assume

$$V_{OH} = 11.5 \text{ V}$$

$$V_{OL} = 0.2 \text{ V}, \beta_F = 25$$

Ans:
a) $Y = \overline{A + B \cdot C}$
b) 13
c) 65.7 mW

Exercise 8



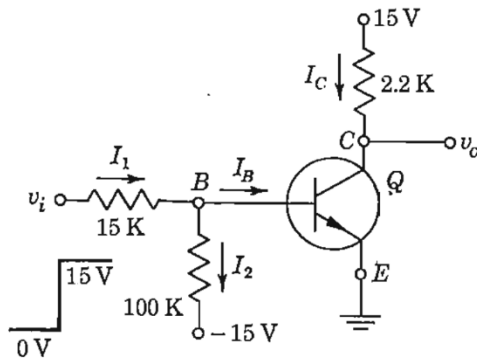
In this circuit $R_1 = R_2 = R_3 = 100\text{ k}\Omega$, $R_C = 10\text{ k}\Omega$, $V_{CC} = 20\text{ V}$.

Now if the input voltages are $V_A = 0\text{ V}$, $V_B = 10\text{ V}$, $V_C = 10\text{ V}$.

- Find out the output voltage.
- Find out the value of I .
- Find out the value of I_2 .
- Find out the value of I_3 .

Ans: a) 0.2 V
 b) 1.98 mA
 c) 0.99 mA
 d) 0.99 mA

Exercise 9



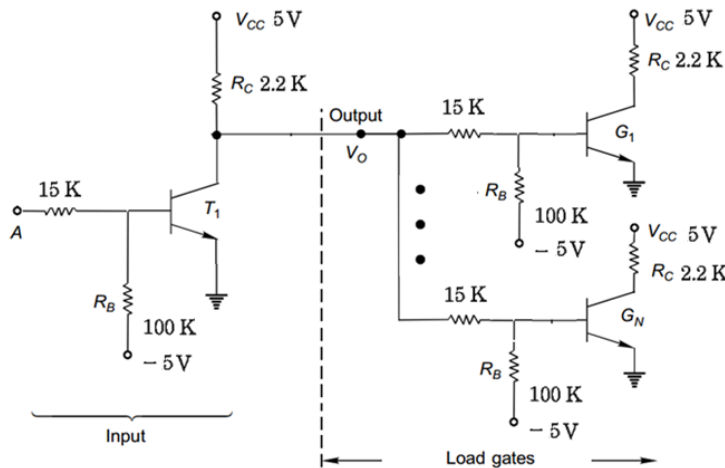
For the RTL NOT gate shown,

When input is high-

- Find the output voltage.
- Find the value of I_C .
- Find the value of I_1 .
- Find the value of I_2 .
- Find the value of I_B .

Ans: a) 0.2 V
 b) 6.7272 mA
 c) 0.94667 mA
 d) 0.158 mA
 e) 0.78867 mA

Exercise 10



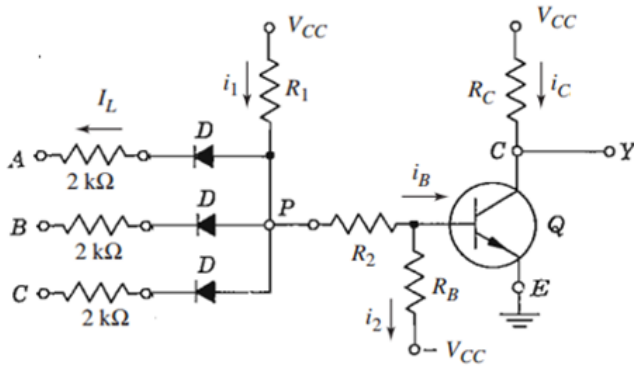
For the above RTL inverter driver & loads, assume

$$V_{OH} = 4\text{ V}, V_{OL} = 0.2\text{ V}, \beta_F = 30$$

- Find the value of V_{IL} .
- Find the value of V_{IH} .
- Calculate the noise margin.
- Find the maximum number of FANOUT for this circuit.

Ans: a) 1.325 V
 b) 2.761 V
 c) 1.125 V
 d) 2

Exercise 11



For the given circuit, $\beta_F = 30$, $V_{CC} = 12\text{ V}$,

$R_1 = R_2 = 15\text{ k}\Omega$, $R_C = 2.2\text{ k}\Omega$, $R_B = 100\text{ k}\Omega$

Assume $V_A = 0.1\text{ V}$, $V_B = V_C = 12\text{ V}$

For this part assume inputs of the load devices are not connected to the driver device.

- Find I_L .
- Find i_1 .
- Find i_2 .
- Find I_B .
- Find I_C .
- Find the power dissipated in this case.

Ans: a) 0.5521 mA
 b) 0.673 mA
 c) 0.121 mA
 d) 0 mA
 e) 0 mA
 f) 9.474 mW

Exercise 12

For the circuit in Exercise 11, assume

$$V_A = V_B = V_C = 12\text{ V}$$

For this part assume inputs of the load devices are not connected to driver device.

- a) Find I_L .
- b) Find i_1 .
- c) Find i_2 .
- d) Find I_B .
- e) Find I_C .
- f) Find the power dissipated in this case.
- g) Find the value of β_{min} .

Ans: a) 0
b) 0.37334 mA
c) 0.128 mA
d) 0.245334 mA
e) 5.41 mA
f) 70.936 mW
g) 22.051